

Math 21 Curriculum

Algebra and its Functions | Middlesex School Mathematics | Textbook: OpenStax Intermediate Algebra 2e (Marecek and Mathis, 2020)

COURSE AT A GLANCE

Review Content

REVIEW

3.2 Slope of a Line; 3.1 Graph Linear Equations in Two Variables; 3.3 Find the Equation of a Line; 5.2 Properties of Exponents and Scientific Notation

UNIT LEARNING OBJECTIVES

- I can determine and interpret the slope of a line.
- I can graph a line from an equation and write an equation from given information.
- I can apply a linear function to a context.
- I can identify the solution of a linear system as an intersection point.
- I can solve a simple two-variable linear system using an appropriate method.
- I can simplify an expression using basic positive-integer exponent rules.

Functions and Function Interpretation

REQUIRED

3.5 Relations and Functions; 3.6 Graphs of Functions

UNIT LEARNING OBJECTIVES

- I can evaluate a function from a formula, graph, or table.
- I can determine whether a relation is a function.
- I can determine domain and range from a graph or table.
- I can solve for an input that produces a specified output.
- I can interpret inputs, outputs, domain, and range in context.
- I can calculate and interpret average rate of change.
- I can compare functions represented by formulas, graphs, or tables.

Polynomials and Factoring

REQUIRED

5.1 Add and Subtract Polynomials; 5.4 Dividing Polynomials; 5.3 Multiply Polynomials; 6.1 Greatest Common Factor and Factor by Grouping; 6.2 Factor Trinomials; 6.3 Factor Special Products

UNIT LEARNING OBJECTIVES

- I can add and subtract polynomials.
- I can multiply polynomials.
- I can factor out a greatest common factor.
- I can factor a quadratic trinomial.
- I can factor a difference of squares and other required special cases.
- I can use factoring to determine the zeros of a polynomial.

Rational Expressions and Equations

REQUIRED

7.1 Simplify Rational Expressions; 7.2 Multiply and Divide Rational Expressions; 7.3 Add and Subtract Rational Expressions with a Common Denominator; 7.4 Add and Subtract Rational Expressions with Unlike Denominators; 7.5 Solve Rational Equations

UNIT LEARNING OBJECTIVES

- I can state excluded values for a rational expression.
- I can simplify a rational expression while preserving its restrictions.
- I can multiply and divide rational expressions.
- I can add and subtract rational expressions using a common denominator.
- I can solve a rational equation and reject extraneous solutions.

Exponents and Radicals

REQUIRED

8.3 Simplify Rational Exponents; 8.1 Simplify Expressions with Roots; 8.2 Simplify Radical Expressions; 8.4 Add, Subtract, and Multiply Radical Expressions

UNIT LEARNING OBJECTIVES

- I can simplify expressions containing negative exponents.
- I can rewrite expressions between radical and rational-exponent forms.
- I can simplify radical expressions.
- I can apply exponent rules to expressions with integer and rational exponents.
- I can solve a basic equation involving a radical or rational exponent and check for extraneous solutions.

Quadratic Functions and Equations

REQUIRED

9.6 Graph Quadratic Functions Using Properties; 9.7 Graph Quadratic Functions Using Transformations; 9.1 Solve Quadratic Equations Using the Square Root Property; 9.2 Solve Quadratic Equations by Completing the Square; 9.3 Solve Quadratic Equations Using the Quadratic Formula; 9.4 Solve Equations in Quadratic Form; 9.5 Solve Applications of Quadratic Equations

UNIT LEARNING OBJECTIVES

- I can identify key features of a quadratic function from its graph or equation.
- I can connect standard, factored, and vertex forms to features of a parabola.
- I can solve a quadratic equation by factoring.
- I can solve a quadratic equation using the square-root property.
- I can solve a quadratic equation by completing the square.
- I can solve a quadratic equation using the quadratic formula.
- I can choose an efficient method for solving a quadratic equation.
- I can graph a quadratic function using its vertex, intercepts, and symmetry.
- I can create and interpret a quadratic model for a projectile, area, or optimization problem.

Extension Content

EXTENSION

9.3 Solve Quadratic Equations Using the Quadratic Formula

UNIT LEARNING OBJECTIVES

- I can perform basic arithmetic with complex numbers.
- I can express the nonreal solutions of a quadratic equation using complex numbers.

IN-DEPTH CURRICULUM

SKILL PROGRESSION



Introduce



Reinforce



Deepen



Apply

Review Content

REVIEW

Textbook coverage: 3.2 Slope of a Line; 3.1 Graph Linear Equations in Two Variables; 3.3 Find the Equation of a Line; 5.2 Properties of Exponents and Scientific Notation

M21-REV-001

I can determine and interpret the slope of a line.

Textbook: 3.2 Slope of a Line

SUPPORTING SKILLS



Determine slope from a graph.

SK-LIN-SLOPE-GRAPH · first introduced in Math 12



Calculate slope from two points.

SK-LIN-SLOPE-POINTS · first introduced in Math 12



Attach and interpret units for rates, inputs, and outputs.

SK-REP-UNITS · first introduced in Math 12

M21-REV-002

I can graph a line from an equation and write an equation from given information.

Textbook: 3.1 Graph Linear Equations in Two Variables; 3.3 Find the Equation of a Line

SUPPORTING SKILLS



Convert among common forms of a linear equation.

SK-LIN-FORM-CONVERT · first introduced in Math 12



Use slope to generate additional points on a line.

SK-LIN-GRAPH-SLOPE · first introduced in Math 12



Plot an intercept accurately on a coordinate plane.

SK-LIN-PLOT-INTERCEPT · first introduced in Math 12



Calculate slope from two points.

SK-LIN-SLOPE-POINTS · first introduced in Math 12

M21-REV-003

I can apply a linear function to a context.

Textbook: 3.3 Find the Equation of a Line

SUPPORTING SKILLS

- I** Explain a model parameter in the language of its context.
SK-MODEL-PARAMETER-INTERPRET
- R** Restrict a model to meaningful contextual inputs.
SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- A** Attach and interpret units for rates, inputs, and outputs.
SK-REP-UNITS · first introduced in Math 12

M21-REV-004

I can identify the solution of a linear system as an intersection point.

Textbook: No mapped textbook section

SUPPORTING SKILLS

- R** Locate and interpret the intersection of two lines.
SK-SYS-GRAPH-INTERSECTION · first introduced in Math 12

M21-REV-005

I can solve a simple two-variable linear system using an appropriate method.

Textbook: No mapped textbook section

SUPPORTING SKILLS

- R** Check an ordered pair in both equations of a system.
SK-SYS-CHECK · first introduced in Math 12
- R** Add equations to eliminate a variable.
SK-SYS-ELIMINATE · first introduced in Math 12
- R** Locate and interpret the intersection of two lines.
SK-SYS-GRAPH-INTERSECTION · first introduced in Math 12
- R** Substitute an equivalent expression for a variable.
SK-SYS-SUBSTITUTE · first introduced in Math 12

M21-REV-006

I can simplify an expression using basic positive-integer exponent rules.

Textbook: 5.2 Properties of Exponents and Scientific Notation

SUPPORTING SKILLS

- R** Apply the power rule for exponents.
SK-EXP-POWER · first introduced in Math 12
- R** Apply the product rule for exponents.
SK-EXP-PRODUCT · first introduced in Math 12
- R** Apply the quotient rule for exponents.
SK-EXP-QUOTIENT · first introduced in Math 12

Functions and Function Interpretation

REQUIRED

Textbook coverage: 3.5 Relations and Functions; 3.6 Graphs of Functions

M21-FUN-001

I can evaluate a function from a formula, graph, or table.

Textbook: 3.5 Relations and Functions; 3.6 Graphs of Functions

SUPPORTING SKILLS

- A** Simplify an algebraic expression using operation properties.
SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- D** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- A** Identify the input and output represented by function notation.
SK-FUNC-NOTATION-READ · first introduced in Math 12

M21-FUN-002

I can determine whether a relation is a function.

Textbook: 3.5 Relations and Functions; 3.6 Graphs of Functions

SUPPORTING SKILLS

- D** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- D** Apply the vertical line test to a graph.
SK-FUNC-VERTICAL-LINE · first introduced in Math 12

M21-FUN-003

I can determine domain and range from a graph or table.

Textbook: 3.5 Relations and Functions; 3.6 Graphs of Functions

SUPPORTING SKILLS

- D** Read domain and range from a graph.
SK-FUNC-DOMAIN-RANGE-GRAPH · first introduced in Math 12
- D** Read domain and range from a table.
SK-FUNC-DOMAIN-RANGE-TABLE · first introduced in Math 12

M21-FUN-004

I can solve for an input that produces a specified output.

Textbook: 3.5 Relations and Functions; 3.6 Graphs of Functions

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I** Solve for an input associated with a specified output.
SK-FUNC-SOLVE-INPUT

M21-FUN-005

I can interpret inputs, outputs, domain, and range in context.

Textbook: 3.5 Relations and Functions; 3.6 Graphs of Functions

SUPPORTING SKILLS

- D** Restrict domain to values meaningful in a context.
SK-FUNC-CONTEXT-DOMAIN · first introduced in Math 12
- D** Interpret the range of a function in a context.
SK-FUNC-CONTEXT-RANGE · first introduced in Math 12
- D** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- D** Restrict a model to meaningful contextual inputs.
SK-MODEL-VALID-DOMAIN · first introduced in Math 12

M21-FUN-006

I can calculate and interpret average rate of change.

Textbook: No mapped textbook section

SUPPORTING SKILLS

- I** Calculate average rate of change over an interval.
SK-FUNC-AROC
- I** Interpret average rate of change with contextual units.
SK-FUNC-AROC-INTERPRET
- D** Attach and interpret units for rates, inputs, and outputs.
SK-REP-UNITS · first introduced in Math 12

M21-FUN-007

I can compare functions represented by formulas, graphs, or tables.

Textbook: No mapped textbook section

SUPPORTING SKILLS

- I** Compare functions shown in different representations.
SK-FUNC-COMPARE-REP
- I** Connect parameters in a formula to features of its graph.
SK-REP-FORMULA-GRAPH
- I** Use tabular patterns to predict graph shape and behavior.
SK-REP-TABLE-GRAPH

Polynomials and Factoring

REQUIRED

Textbook coverage: 5.1 Add and Subtract Polynomials; 5.4 Dividing Polynomials; 5.3 Multiply Polynomials; 6.1 Greatest Common Factor and Factor by Grouping; 6.2 Factor Trinomials; 6.3 Factor Special Products

M21-POL-001

I can add and subtract polynomials.

Textbook: 5.1 Add and Subtract Polynomials; 5.4 Dividing Polynomials

SUPPORTING SKILLS

- A** Identify and combine like terms.
SK-ALG-LIKE-TERMS · inherited from middle school
- D** Identify like polynomial terms by variable part and exponent.
SK-POLY-LIKE-TERMS · first introduced in Math 12
- R** Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
SK-POLY-VOCABULARY · first introduced in Math 12

M21-POL-002

I can multiply polynomials.

Textbook: 5.3 Multiply Polynomials; 5.4 Dividing Polynomials

SUPPORTING SKILLS

- A** Apply the distributive property in either direction.
SK-ALG-DISTRIBUTE · inherited from middle school
- D** Multiply two binomials using distribution.
SK-POLY-BINOMIAL-MULTIPLY · first introduced in Math 12
- D** Multiply a monomial by a polynomial.
SK-POLY-DISTRIBUTE · first introduced in Math 12
- A** Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
SK-POLY-VOCABULARY · first introduced in Math 12

M21-POL-003

I can factor out a greatest common factor.

Textbook: 6.1 Greatest Common Factor and Factor by Grouping

SUPPORTING SKILLS

- R** Verify a factorization by multiplication.
SK-FACTOR-CHECK · first introduced in Math 12
- D** Factor the greatest common monomial factor from a polynomial.
SK-FACTOR-GCF · first introduced in Math 12
- A** Determine the numerical and variable greatest common factor of polynomial terms.
SK-POLY-GCF · first introduced in Math 12
- A** Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
SK-POLY-VOCABULARY · first introduced in Math 12

M21-POL-004

I can factor a quadratic trinomial.

Textbook: 6.1 Greatest Common Factor and Factor by Grouping; 6.2 Factor Trinomials

SUPPORTING SKILLS

- A** Verify a factorization by multiplication.
SK-FACTOR-CHECK · first introduced in Math 12
- R** Factor a quadratic trinomial with a nonunit leading coefficient.
SK-FACTOR-TRINOMIAL-GENERAL · first introduced in Math 12
- R** Factor a monic quadratic trinomial.
SK-FACTOR-TRINOMIAL-MONIC · first introduced in Math 12
- A** Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
SK-POLY-VOCABULARY · first introduced in Math 12

M21-POL-005

I can factor a difference of squares and other required special cases.

Textbook: 6.3 Factor Special Products

SUPPORTING SKILLS

- A** Verify a factorization by multiplication.
SK-FACTOR-CHECK · first introduced in Math 12
- I** Factor a difference of two squares.
SK-FACTOR-DIFF-SQUARES
- A** Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
SK-POLY-VOCABULARY · first introduced in Math 12

M21-POL-006

I can use factoring to determine the zeros of a polynomial.

Textbook: No mapped textbook section

SUPPORTING SKILLS

- A** Factor a difference of two squares.
SK-FACTOR-DIFF-SQUARES · first introduced in Math 21
- A** Factor a quadratic trinomial with a nonunit leading coefficient.
SK-FACTOR-TRINOMIAL-GENERAL · first introduced in Math 12
- A** Factor a monic quadratic trinomial.
SK-FACTOR-TRINOMIAL-MONIC · first introduced in Math 12
- I** Apply the zero-product property.
SK-FACTOR-ZERO-PRODUCT
- A** Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
SK-POLY-VOCABULARY · first introduced in Math 12

Rational Expressions and Equations

REQUIRED

Textbook coverage: 7.1 Simplify Rational Expressions; 7.2 Multiply and Divide Rational Expressions; 7.3 Add and Subtract Rational Expressions with a Common Denominator; 7.4 Add and Subtract Rational Expressions with Unlike Denominators; 7.5 Solve Rational Equations

M21-RAT-001

I can state excluded values for a rational expression.

Textbook: 7.1 Simplify Rational Expressions; 7.2 Multiply and Divide Rational Expressions; 7.3 Add and Subtract Rational Expressions with a Common Denominator; 7.4 Add and Subtract Rational Expressions with Unlike Denominators

SUPPORTING SKILLS

- I** Determine excluded values from an original denominator.
SK-RAT-RESTRICTIONS
 - I** Recall and correctly use rational-expression terms including numerator, denominator, restriction, and excluded value.
SK-RAT-VOCABULARY
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M21-RAT-002

I can simplify a rational expression while preserving its restrictions.

Textbook: 7.1 Simplify Rational Expressions; 7.2 Multiply and Divide Rational Expressions

SUPPORTING SKILLS

- I** Cancel common factors rather than individual terms.
SK-RAT-CANCEL-FACTORS
 - I** Factor numerators and denominators completely.
SK-RAT-FACTOR
 - A** Determine excluded values from an original denominator.
SK-RAT-RESTRICTIONS · first introduced in Math 21
 - A** Recall and correctly use rational-expression terms including numerator, denominator, restriction, and excluded value.
SK-RAT-VOCABULARY · first introduced in Math 21
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M21-RAT-003

I can multiply and divide rational expressions.

Textbook: 7.2 Multiply and Divide Rational Expressions

SUPPORTING SKILLS

- A** Add, subtract, multiply, and divide fractions.
SK-NUM-FRACTION-OPERATE · inherited from middle school
- A** Cancel common factors rather than individual terms.
SK-RAT-CANCEL-FACTORS · first introduced in Math 21
- A** Factor numerators and denominators completely.
SK-RAT-FACTOR · first introduced in Math 21
- A** Determine excluded values from an original denominator.
SK-RAT-RESTRICTIONS · first introduced in Math 21

M21-RAT-004

I can add and subtract rational expressions using a common denominator.

Textbook: 7.3 Add and Subtract Rational Expressions with a Common Denominator; 7.4 Add and Subtract Rational Expressions with Unlike Denominators

SUPPORTING SKILLS

- A** Add, subtract, multiply, and divide fractions.
SK-NUM-FRACTION-OPERATE · inherited from middle school
- I** Combine rational expressions over a common denominator.
SK-RAT-COMBINE
- I** Determine the least common denominator of rational expressions.
SK-RAT-LCD

M21-RAT-005

I can solve a rational equation and reject extraneous solutions.

Textbook: 7.5 Solve Rational Equations

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I** Clear denominators from a rational equation.
SK-RAT-CLEAR-DENOMINATORS
- I** Check candidate solutions against denominator restrictions.
SK-RAT-EXTRANEIOUS
- A** Determine excluded values from an original denominator.
SK-RAT-RESTRICTIONS · first introduced in Math 21

Exponents and Radicals

REQUIRED

Textbook coverage: 8.3 Simplify Rational Exponents; 8.1 Simplify Expressions with Roots; 8.2 Simplify Radical Expressions; 8.4 Add, Subtract, and Multiply Radical Expressions

M21-EXP-001

I can simplify expressions containing negative exponents.

Textbook: 8.3 Simplify Rational Exponents

SUPPORTING SKILLS

- I** Rewrite a negative exponent using a reciprocal.
SK-EXP-NEGATIVE
- A** Apply the product rule for exponents.
SK-EXP-PRODUCT · first introduced in Math 12
- A** Apply the quotient rule for exponents.
SK-EXP-QUOTIENT · first introduced in Math 12

M21-EXP-002

I can rewrite expressions between radical and rational-exponent forms.

Textbook: 8.3 Simplify Rational Exponents

SUPPORTING SKILLS

- I** Interpret numerator and denominator roles in a rational exponent.
SK-EXP-RATIONAL
- I** Convert between radical and rational-exponent notation.
SK-RAD-CONVERT
- I** Read radical and rational-exponent notation and identify the index, radicand, base, numerator, and denominator.
SK-RAD-NOTATION-READ

M21-EXP-003

I can simplify radical expressions.

Textbook: 8.1 Simplify Expressions with Roots; 8.2 Simplify Radical Expressions; 8.4 Add, Subtract, and Multiply Radical Expressions

SUPPORTING SKILLS

- A** Express a whole number as a product of prime factors.
SK-NUM-PRIME-FACTOR · inherited from middle school
- I** Extract perfect-power factors from a radical.
SK-RAD-FACTOR-PERFECT

M21-EXP-004

I can apply exponent rules to expressions with integer and rational exponents.

Textbook: 8.3 Simplify Rational Exponents; 8.4 Add, Subtract, and Multiply Radical Expressions

SUPPORTING SKILLS

- A** Rewrite a negative exponent using a reciprocal.
SK-EXP-NEGATIVE · first introduced in Math 21
- D** Apply the power rule for exponents.
SK-EXP-POWER · first introduced in Math 12
- D** Apply the product rule for exponents.
SK-EXP-PRODUCT · first introduced in Math 12
- D** Apply the quotient rule for exponents.
SK-EXP-QUOTIENT · first introduced in Math 12
- A** Interpret numerator and denominator roles in a rational exponent.
SK-EXP-RATIONAL · first introduced in Math 21
- I** Apply the zero-exponent rule.
SK-EXP-ZERO

M21-EXP-005

I can solve a basic equation involving a radical or rational exponent and check for extraneous solutions.

Textbook: No mapped textbook section

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I** Require an even-root radicand to be nonnegative when determining domain.
SK-FUNC-DOMAIN-EVEN-ROOT
- I** Check a candidate solution for extraneous results.
SK-RAD-CHECK
- I** Isolate a radical expression before solving.
SK-RAD-ISOLATE

Quadratic Functions and Equations

REQUIRED

Textbook coverage: 9.6 Graph Quadratic Functions Using Properties; 9.7 Graph Quadratic Functions Using Transformations; 9.1 Solve Quadratic Equations Using the Square Root Property; 9.2 Solve Quadratic Equations by Completing the Square; 9.3 Solve Quadratic Equations Using the Quadratic Formula; 9.4 Solve Equations in Quadratic Form; 9.5 Solve Applications of Quadratic Equations

M21-QUAD-001

I can identify key features of a quadratic function from its graph or equation.

Textbook: 9.6 Graph Quadratic Functions Using Properties; 9.7 Graph Quadratic Functions Using Transformations

SUPPORTING SKILLS

- I** Identify standard, factored, and vertex forms.
SK-QUAD-FORM-IDENTIFY
- I** Determine x- and y-intercepts of a quadratic function.
SK-QUAD-INTERCEPTS
- I** Determine the vertex and axis of symmetry.
SK-QUAD-VERTEX
- I** Recall and correctly use quadratic-function terms including parabola, vertex, axis of symmetry, intercept, zero, and solution.
SK-QUAD-VOCABULARY

M21-QUAD-002

I can connect standard, factored, and vertex forms to features of a parabola.

Textbook: 9.6 Graph Quadratic Functions Using Properties; 9.7 Graph Quadratic Functions Using Transformations

SUPPORTING SKILLS

- I** Rewrite a quadratic among standard, factored, and vertex forms using an appropriate algebraic method.
SK-QUAD-FORM-CONVERT
- D** Identify standard, factored, and vertex forms.
SK-QUAD-FORM-IDENTIFY · first introduced in Math 21
- A** Determine x- and y-intercepts of a quadratic function.
SK-QUAD-INTERCEPTS · first introduced in Math 21
- A** Determine the vertex and axis of symmetry.
SK-QUAD-VERTEX · first introduced in Math 21
- D** Connect parameters in a formula to features of its graph.
SK-REP-FORMULA-GRAPH · first introduced in Math 21

M21-QUAD-003

I can solve a quadratic equation by factoring.

Textbook: No mapped textbook section

SUPPORTING SKILLS

- A** Factor a quadratic trinomial with a nonunit leading coefficient.
SK-FACTOR-TRINOMIAL-GENERAL · first introduced in Math 12
- A** Factor a monic quadratic trinomial.
SK-FACTOR-TRINOMIAL-MONIC · first introduced in Math 12
- D** Apply the zero-product property.
SK-FACTOR-ZERO-PRODUCT · first introduced in Math 21
- I** Determine real and complex zeros using an appropriate method.
SK-POLY-ZEROS

M21-QUAD-004

I can solve a quadratic equation using the square-root property.

Textbook: 9.1 Solve Quadratic Equations Using the Square Root Property

SUPPORTING SKILLS

- I** Apply the square-root property with both signs.
SK-QUAD-SQUARE-ROOT
- A** Extract perfect-power factors from a radical.
SK-RAD-FACTOR-PERFECT · first introduced in Math 21

M21-QUAD-005

I can solve a quadratic equation by completing the square.

Textbook: 9.2 Solve Quadratic Equations by Completing the Square

SUPPORTING SKILLS

- A** Simplify an algebraic expression using operation properties.
SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- I** Create a perfect-square trinomial by completing the square.
SK-QUAD-COMPLETE-SQUARE

M21-QUAD-006

I can solve a quadratic equation using the quadratic formula.

Textbook: 9.3 Solve Quadratic Equations Using the Quadratic Formula

SUPPORTING SKILLS

- I** Use the discriminant to predict the number and type of solutions.
SK-QUAD-DISCRIMINANT
- I** Substitute coefficients accurately into the quadratic formula.
SK-QUAD-FORMULA
- A** Extract perfect-power factors from a radical.
SK-RAD-FACTOR-PERFECT · first introduced in Math 21

M21-QUAD-007

I can choose an efficient method for solving a quadratic equation.

Textbook: 9.1 Solve Quadratic Equations Using the Square Root Property; 9.2 Solve Quadratic Equations by Completing the Square; 9.3 Solve Quadratic Equations Using the Quadratic Formula; 9.4 Solve Equations in Quadratic Form

SUPPORTING SKILLS

- A** Apply the zero-product property.
SK-FACTOR-ZERO-PRODUCT · first introduced in Math 21
- A** Create a perfect-square trinomial by completing the square.
SK-QUAD-COMPLETE-SQUARE · first introduced in Math 21
- A** Substitute coefficients accurately into the quadratic formula.
SK-QUAD-FORMULA · first introduced in Math 21
- I** Select an efficient quadratic-solving method from equation structure.
SK-QUAD-METHOD-SELECT
- A** Apply the square-root property with both signs.
SK-QUAD-SQUARE-ROOT · first introduced in Math 21

M21-QUAD-008

I can graph a quadratic function using its vertex, intercepts, and symmetry.

Textbook: 9.6 Graph Quadratic Functions Using Properties; 9.7 Graph Quadratic Functions Using Transformations

SUPPORTING SKILLS

- A** Rewrite a quadratic among standard, factored, and vertex forms using an appropriate algebraic method.
SK-QUAD-FORM-CONVERT · first introduced in Math 21
- D** Determine x- and y-intercepts of a quadratic function.
SK-QUAD-INTERCEPTS · first introduced in Math 21
- D** Determine the vertex and axis of symmetry.
SK-QUAD-VERTEX · first introduced in Math 21
- D** Connect parameters in a formula to features of its graph.
SK-REP-FORMULA-GRAPH · first introduced in Math 21

M21-QUAD-009

I can create and interpret a quadratic model for a projectile, area, or optimization problem.

Textbook: 9.5 Solve Applications of Quadratic Equations

SUPPORTING SKILLS

- D** Explain a model parameter in the language of its context.
SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A** Restrict a model to meaningful contextual inputs.
SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- I** Interpret vertex and zeros in an application.
SK-QUAD-MODEL-FEATURES
- A** Attach and interpret units for rates, inputs, and outputs.
SK-REP-UNITS · first introduced in Math 12

Extension Content

EXTENSION

Textbook coverage: 9.3 Solve Quadratic Equations Using the Quadratic Formula

M21-EXT-001

I can perform basic arithmetic with complex numbers.

Textbook: No mapped textbook section

SUPPORTING SKILLS

- I** Simplify integer powers of the imaginary unit.
SK-CPLX-I-POWERS
 - I** Add, subtract, and multiply complex numbers.
SK-CPLX-OPERATE
-

M21-EXT-002

I can express the nonreal solutions of a quadratic equation using complex numbers.

Textbook: 9.3 Solve Quadratic Equations Using the Quadratic Formula

SUPPORTING SKILLS

- A** Simplify integer powers of the imaginary unit.
SK-CPLX-I-POWERS · first introduced in Math 21
- I** Express a quadratic solution with a negative discriminant in complex form.
SK-CPLX-QUAD-SOLUTION
- A** Use the discriminant to predict the number and type of solutions.
SK-QUAD-DISCRIMINANT · first introduced in Math 21